ELEC60029 Semiconductor Devices- 2024: *Answers:*

Q1

(a)

1st term: drift – This is the movement of carriers (here, electrons at concentration $n(x)$) under influence of an electric field E. The movement is hindered by scattering processes that limit the final velocity of the carriers. For low electric fields, the carrier velocity varies linearly with electric field, with the proportionality constant corresponding to the mobility μ. Hence, the expression is only valid for low electric field.

[Answer should describe the meaning of the constituent terms, and discuss the limits of the equation]

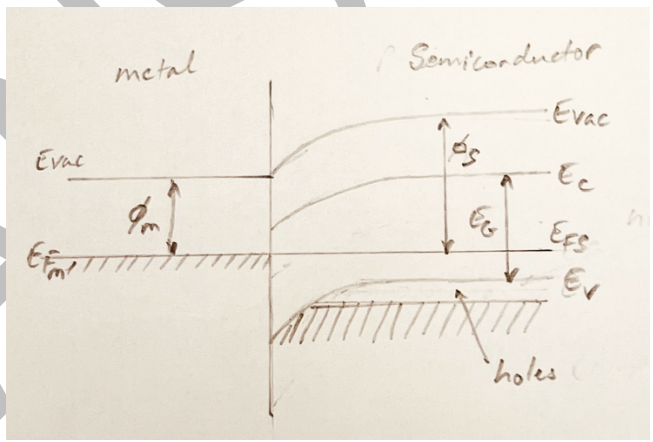
2nd term: diffusion – This is the movement of carriers (here electrons) under influence of a carrier gradient, given by the first derivative of the carrier concentration $n(x)$. Carriers diffuse from higher to lower concentrations via a 'random walk' associated with their thermal energy kT. D_e is the Einstein diffusion constant. The sign of the term gives the correct direction of electron current density, e.g. a reducing gradient of $n(x)$ implies electron flow towards increasing x, and hence a negative current density direction.

[Answer should describe the meaning of the constituent terms, and comment on the physical meaning of diffusion]

[4]

(b)

The band diagram for a metal to p-type semiconductor contact is shown below, where the work function in the metal $\phi_m < \phi_s$, the work-function in the semiconductor. Upon contact the Fermi levels align $E_{Fs} = E_{Fm}$ and a contact potential difference occurs across the junction. The downward bending of E_V produces a potential barrier in the valence band, preventing the flow of holes from the semiconductor into the metal and electrons from the metal into empty states (holes) in the semiconductor. Note that the opposite situation $\phi_m > \phi_s$ would lead to an Ohmic contact.



[4]

(c)

The volume element is shown in the figure, below right. The element has a width Δx and constant cross-sectional area A. Application of Gauss's Law $\int_S \boldsymbol{E}.\mathrm{d}\boldsymbol{A} = \frac{\rho(x)}{\epsilon}$ is straightforward, with the dot products between $\boldsymbol{E}$ and $\mathrm{d}\boldsymbol{A}$ zero for the upper, lower, front and back side surfaces, due to the 90^0 angle between $\boldsymbol{E}$ and $\mathrm{d}\boldsymbol{A}$, and non-zero for the left and right surfaces, due to zero angle between $\boldsymbol{E}$ and $\mathrm{d}\boldsymbol{A}$. This gives:

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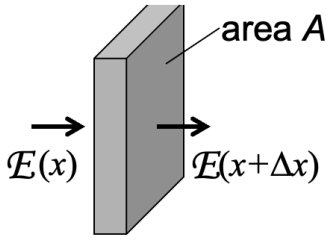
$$\varepsilon \times \mathcal{E}(x+\Delta x)A - \varepsilon \times \mathcal{E}(x)A = \rho(x)A\Delta x$$

$$\frac{\mathcal{E}(x+\Delta x)-\mathcal{E}(x)}{\Delta x} = \frac{\rho(x)}{\varepsilon}$$

$$\frac{\mathcal{E}(x)-\mathcal{E}(x+\Delta x)}{x-(x+\Delta x)} = \frac{\rho(x)}{\varepsilon}$$

$$\lim_{\Delta x \to 0}$$

$$\frac{d\mathcal{E}(x)}{dx} = \frac{\rho(x)}{\varepsilon}$$



ε: permittivity (F/cm)

ρ: charge density (C/cm^3)Δx

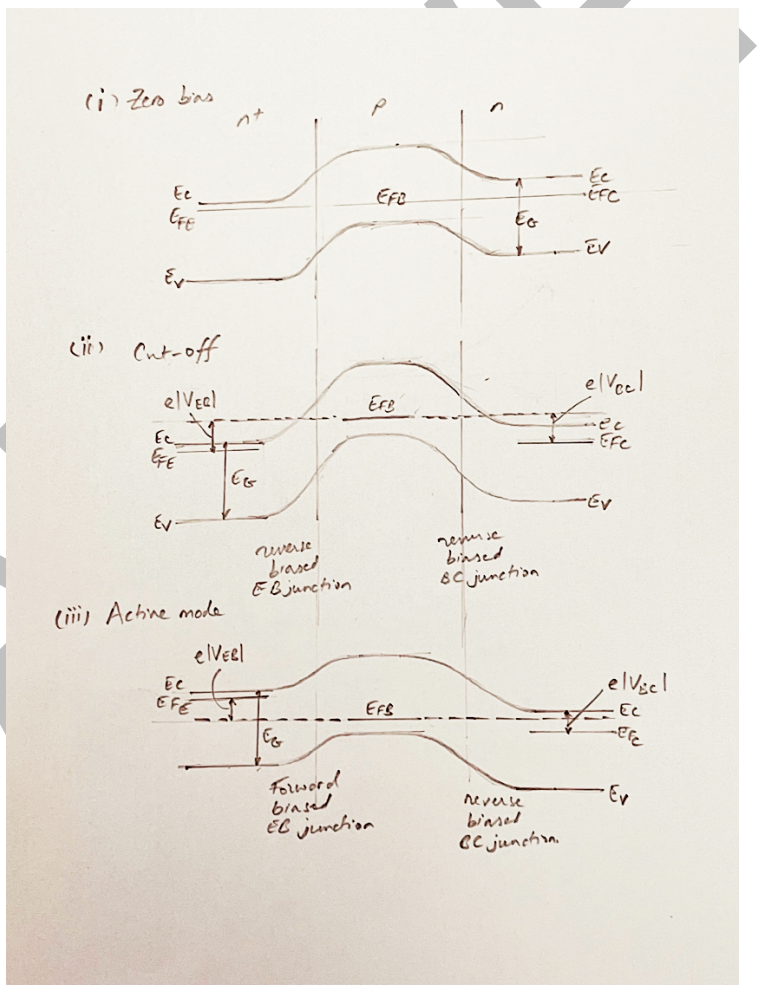
[4]

This is the general form of Poisson's Equation. For a semiconductor, $\rho(x)$ can have contributions from mobile (electrons and holes) and fixed (ionised dopant atoms) charges. This gives the following form:

$$\varepsilon \frac{d\mathcal{E}(x)}{dx} = \rho(x) = e\left(-n(x) + p(x) + N_D^+(x) - N_A^-(x)\right)$$

[2]

(d)



[4 marks per band diagram]

(e)

i) $V_{GS1} < V_{GS2} < V_{GS3}$ [2]

ii) '1' is the triode region where $V_{DS} << V_{GS}-V_{th}$. Expression of MOSFET current from formulae sheet:

$$I_{DS} = \frac{\mu C_{ox} W}{L}\left((V_{GS} - V_{th})V_{DS} - \frac{V_{DS}^2}{2}\right)$$

For $V_{DS} << V_{GS}-V_{th}$, $\left((V_{GS} - V_{th})V_{DS} >> \frac{V_{DS}^2}{2}\right)$ thus $\left(\frac{V_{DS}^2}{2}\right)$ negligible compared to $\left((V_{GS} - V_{th})V_{DS}\right)$, therefore:

$$I_{DS} \approx \frac{\mu C_{ox} W}{L}\left((V_{GS} - V_{th})V_{DS}\right)$$ [4]

iii) $V_{DS} > V_{GS}-V_{th}$ is the saturation region. Expression of MOSFET current from formulae sheet:

$$I_{DS} = \frac{\mu C_{ox} W}{L}\left((V_{GS} - V_{th})V_{DS} - \frac{V_{DS}^2}{2}\right)$$

For $V_{DS} > V_{GS}-V_{th}$, the current becomes constant with the same value as the current at $V_{DS} = V_{GS}-V_{th}$, therefore:

$$I_{DS} \approx \frac{\mu C_{ox} W}{2L}(V_{GS} - V_{th})^2 .$$ [4]

iv) On the dashed line $V_{DS}=V_{GS}-V_{th}$ thus we replace $V_{GS}-V_{th}$ by V_{DS} in $I_{DS} = \frac{\mu C_{ox} W}{L}\left((V_{GS} - V_{th})V_{DS} - \frac{V_{DS}^2}{2}\right)$ for the I_{DS} versus V_{DS} characteristics that is the dashed line: $I_{DS} \approx \frac{\mu C_{ox} W}{2L} V_{DS}^{\ 2}$ [4]

ANSWERS

Q2

a)

(i)

Majority carriers are holes, as E_F is closer to E_V.

Using $p = N_v exp\left(\frac{E_v - E_F}{kT}\right) = 1.8 \times 10^{19} exp\left(\frac{-0.2}{0.026}\right) = 8.2 \times 10^{15} cm^{-3}$

Here, the value for N_v is taken from the formula sheet.

[4]

(ii) Using expression for n and p from the formula sheet, substituting $n = n_i$, $p = p_i$ and $E_F = E_i$ via the definition of the intrinsic level:

$n_i = N_c exp\left(-\frac{E_c - E_i}{kT}\right)$ and $p_i = N_v exp\left(-\frac{E_i - E_v}{kT}\right)$

Hence, $p_i n_i = N_c N_v exp\left(-\frac{E_c - E_i + E_i - E_v}{kT}\right) = N_c N_v exp\left(-\frac{E_c - E_v}{kT}\right)$ [2]

$\Longrightarrow p_i n_i = n_i^2 = N_c N_v exp\left(-\frac{E_G}{kT}\right)$

$\Longrightarrow n_i = \sqrt{(N_c N_v)} exp\left(-\frac{E_G}{2kT}\right)$ [2]

Hence, $n_i = \sqrt{(N_c N_v)} exp\left(-\frac{E_G}{2kT}\right) = N_c exp\left(-\frac{E_c - E_i}{kT}\right)$

$\Longrightarrow \sqrt{\frac{N_c}{N_v}} exp\left(\frac{E_G}{2kT}\right) = exp\left(\frac{E_c - E_i}{kT}\right)$

$\Longrightarrow ln\sqrt{\frac{N_c}{N_v}} + \frac{E_G}{2kT} = \left(\frac{E_c - E_i}{kT}\right)$

$\Longrightarrow E_c - E_i = kT ln\sqrt{\frac{N_c}{N_v}} + \frac{E_G}{2} = 0.026 ln\sqrt{\frac{3.2\times10^{19}}{1.8\times10^{19}}} + \frac{1.12}{2}$=0.0075+0.56=0.57eV

This is close to, but not quite, mid-gap

The values for N_c and N_v are taken from the formula sheet.

[4]

(b) (i)

The diode is a p^+n, diode, hence current is dominated by hole injection into the n region, with corresponding hole current. The diode is to be switched from 'off' to 'on', at time $t = 0$. Assuming $V > v_d$, the forward bias diode voltage, the final current after S is closed, is determined by the external circuit, $I = Ip = V/R = Q_p/\tau_p$, where τ_p is the hole recombination lifetime.

[2]

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Hence,

$$@t<0; i_p(0^-)=0; Q_p(0^-)=0$$

$$@t=0^+; i_p(0^+)=I_{ON}=\frac{V}{R}$$

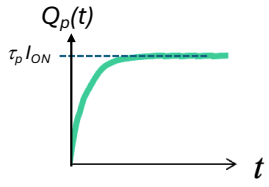
$$I_{ON}=\frac{Q_p(t)}{\tau_p}+\frac{dQ_p(t)}{dt}$$

Solving the differential equation for I_{ON} with the above boundary conditions then gives:

$$Q_p(t)=\tau_p I_{ON}\left(1-\exp\left(-\frac{t}{\tau_p}\right)\right)$$

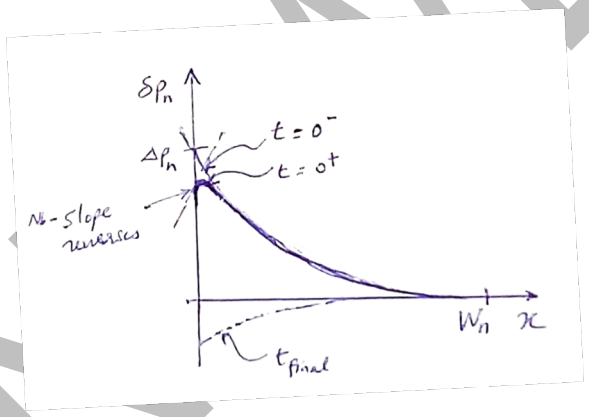
[4]

This is sketched as follows:



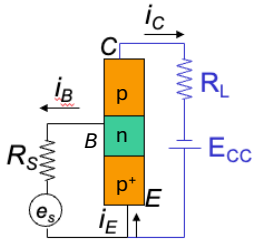
[2]

(ii) Sketches for $t = 0^-$, $t = 0^+$ and the final, steady state, are shown below. Note the reversal in slope for the $t = 0^+$ curve near $x = 0$.



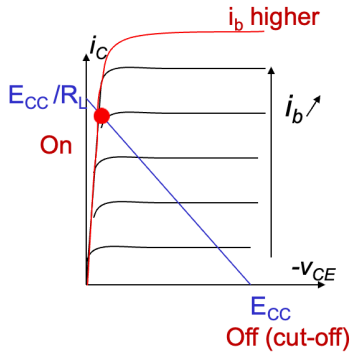
[2+2+2]

(c) The circuit diagram for a p^+np BJT in common-emitter (CE) mode is shown below:



Oversaturation of a BJT in common-emitter (CE) mode (figure below) is an effect where the output current i_C is restricted to $i_C = V_{CC}/R_L$ by the external circuit load-line (blue line) associated with supply voltage V_{CC} and load resistor R_L. The restriction of current is in comparison with the output current $i_C = \beta i_B$ which the BJT attempts to drive via a base current i_B, i.e., $I_{CE} = V_{CC}/R_L < \beta I_B$. This situation is illustrated in the figure below:

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While this leads to greater injected charge build up from the emitter into the base, in this case excess holes, it also leads to a shorter BJT turn-on time, due to a steeper exponential rise in i_c.

[6]

[The equation for $i_c(t)$ is given by:

$$i_C(t) = \frac{Q_B(t)}{\tau_t} = \frac{I_p \tau_p}{\tau_t}\left[1 - \exp\left(\frac{-t}{\tau_p}\right)\right]$$

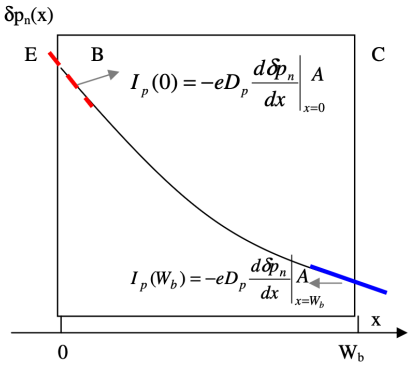
As the current saturates at a lower value, $i_C = V_{CC}/R_L$, than the final saturation current of $I_p \tau_p/\tau_t$ predicted by this equation, the rise in i_c occurs along a steeper part of the exponential curve and hence is quicker.

While candidates are not expected to give the above equation, however they should comment on the implications of oversaturation in switching the BJT 'on', i.e. the faster turn-on time]

ANSWERS

Q3:

(a)



The hole current injected across the EB junction at $x = 0$ is proportional to the slope of the excess minority carrier concentration at $x = 0$. The hole current injected across the BC junction is given by the slope of the minority carrier concentration at $x = W_b$. These slopes are different due to the net curve consisting of a forward and backward decaying exponential, with underlying carrier recombination. The difference between the two hole currents is the recombination current in the base layer.

Comment: [2] for having the minority carrier type correct
[2] relation of current to slope
[4] the correct formulae with the correct slopes and position thereof

[8]

(b) $\gamma = 1$ means that there is no current flow from the base into emitter. The base current is then only composed of recombination re-supply current. In an *npn* BJT, the base current is formed by holes, the collector and emitter current by electrons.

$$I_c = I_n = \frac{Q_n}{\tau_t}$$

$$I_b = \frac{Q_n}{\tau}$$

[5]

Here, Q_n is the minority carrier excess charge (electrons) in the base.
τ_t, is the transit time - the time it takes injected electrons to cross the base from emitter to collector junction.
τ is the minority carrier (electron) lifetime in the base, i.e., the average time minority carriers spend in the base before recombining.

[3]

[Note: If taking I_{CBO} into account here, this current needs to be introduced in both I_c as well as I_b, as shown in the equations below. Note the signs for I_{CBO} and that in forward active mode, this current is negligible.

$I_c = I_n = \frac{Q_n}{\tau_t} + I_{CBO}$ $\quad$ $I_b = \frac{Q_n}{\tau} - I_{CBO}$]

(c) Once the collector and base current can be expressed in simple expressions as given in b), the current gain is given by:

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$$\beta = \frac{I_c}{I_b} = \frac{\frac{Q_n}{\tau_t}}{\frac{Q_n}{\tau}} = \frac{\tau}{\tau_t}$$

[4]

The effective base width will be the metallurgic base width minus the depletion width extending into the base region. The depletion region expressions are given in the formulae sheet:

$$w_p = \sqrt{\frac{2\varepsilon}{e}\frac{N_D}{N_A N_D + N_A^2}(V_0 - V)}$$

Here,

$$V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right) = 0.026V \ln\left(\frac{10^{17}\times 5\times 10^{16}}{(1.45\times 10^{10})^2}\right) = 0.80V$$

$$w_p = \sqrt{\frac{2\varepsilon}{e}\frac{N_D}{N_A N_D + N_A^2}(V_0 - V)} = \sqrt{\frac{2\times 11\times 8.85\times 10^{-14}}{1.6\times 10^{-19}}\frac{5\times 10^{16}}{5\times 10^{16}\times 10^{17} + 10^{34}}(0.8-(-1))}$$

$$w_p = 8.5\times 10^{-6} cm$$

[6]

This implies that the un-depleted base width is:

$$W_B' = 10^{-4} - 8.5\times 10^{-6} = 9.15\times 10^{-5} cm$$

The transit time, using the given expression, is:

$$\tau_t = \frac{W_B^2}{2D_n} = 6.98\times 10^{-10} s$$

Hence, the current gain becomes:

$$\beta = \frac{\tau}{\tau_t} = \frac{10^{-6}}{6.98\times 10^{-10}} = 1433$$

[4]

ANSWERS

Q4:

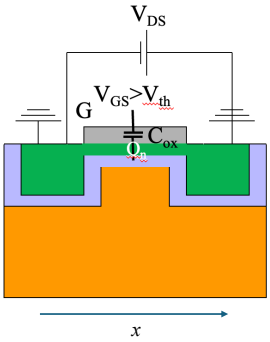
(a)

(i). This is an n-channel enhancement mode MOSFET, hence $V_{th} > 0$ V.

[Marks 2]

(ii)

A sketch of the cross-section of the MOSFET channel region is shown below (The answer should provide a sketch, or describe the situation in text):



[Marks 2]

To derive an expression for the inversion layer charge Q_n as a function of $V(x)$, consider the MOSFET cross-section above. With the channel along x-axis, $x = 0$ at the source end of the channel region, $V_{DS} = 0$, and applying $V_{GS} \geq V_{th}$, the inversion layer charge:

Q_n = charge/unit area $\Rightarrow$ $Q_n = C_{ox} V$, where V is the net voltage across $C_{ox} = e_0 e_{ox}/t_{ox}$.

We now assume that at $V_{GS} = V_{th}$, $Q_n = 0$

This implies that:

$$Q_n = -C_{ox} (V_{GS} - V_{th})$$

[Marks 2]

We now take $V_{DS} > 0$, $V_{GS} \geq V_{th}$

Not only do we now have a voltage in the y direction due to the gate voltage, but also along x due to V_{ds}.

Taking the voltage in the channel due to V_{DS} as $V_x(x)$, and the voltage across the oxide for each point x in the channel as $V_y(x)$:

At $x = 0$ (source end of channel), $V_y(0) = V_{GS}$
At $x = L$ (drain end of channel region), $V_y(L) = V_{GD}$

Furthermore, $V_{GD} = V_{GS} - V_{DS}$

Therefore, using $V_y(x) = V_{GS} - V_x(x)$, the charge in the channel is:

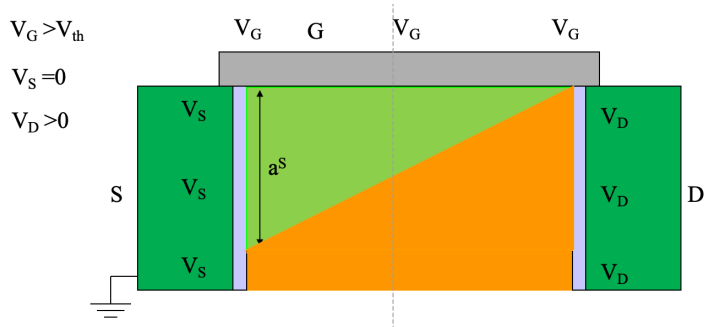
$$Q_n = -C_{ox} (V_{GS} - V_{th} - V_x(x))$$

[Marks 4]

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(b)

Consider the diagram below, assuming a linear variation in voltage $V(x)$ along the channel, and hence channel depth $a(x)$, as we have only just reached pinch-off. This gives the expressions for Q_n and a at the source and drain as given below:



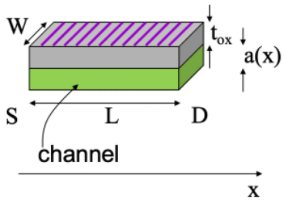
Charge:	$Q_n^S = -C_{ox}(V_{GS} - V_{th})$	$Q_n^D = -C_{ox}((V_{GS} - V_{DS}) - V_{th})$
Inversion layer width:	$a^S = a_0 Q_n^S$	$Q_n^D = -C_{ox}((V_{GS} - (V_{GS} - V_{th})) - V_{th})$ $Q_n^D = 0 \rightarrow a^D = 0$

[Give 2 marks for each pair of expressions on the source and drain side]

[Marks 4]

(c) To find the channel drift current I, given the mobile carrier density $n(x)$, electron mobility μ_e, electric field E and channel cross-sectional area $A = Wa(x)$ for the current (see diagram), we have:

$$I = e\, n(x) \mu_e E\, A$$



[Marks 4]

Furthermore, $Q_n = -C_{ox}\,(V_{GS} - V_x(x) - V_{th})$ = -e × No. of carriers/gate area(=WL), where the term $V_{GS} - V_x(x)$ is the voltage across the oxide at point x.

Therefore, $n(x)$ can be written as:

$$n(x) = \text{no. of carriers/volume} = Q_n(x)/(-ea(x)) = [C_{ox}\,(V_{GS} - V_x(x) - V_{th})]/ea(x)$$

[Marks 4]

This gives:

$$\begin{aligned} I &= [C_{ox}\,(V_{GS} - V_x(x) - V_{th})]\, \mu_e E\, A \,/\, a(x) \\ &= [C_{ox}\,(V_{GS} - V_x(x) - V_{th})]\, \mu_e E\, Wa(x) \,/\, a(x) \\ &= [C_{ox}\,(V_{GS} - V_x(x) - V_{th})]\, \mu_e W\, dV_x/dx \end{aligned}$$

[Marks 4]

ANSWERS

Integrating:

$$I = C_{ox} (V_{GS} - V(x) - V_{th}) W \mu_e dV_x/dx$$

$$\int_0^L I \, dx = C_{ox} W \mu_e \int_0^{V_{DS}} (V_{GS} - V_x(x) - V_{th}) \, dV_x$$

$$I = C_{ox} \mu_e (W/L) [(V_{GS} - V_{th}) V_{DS} - V_{DS}^2/2]$$

[Marks 4]

ANSWERS